

instance rand_n8_m8_d0.3_s11

claim: the minimum is 8 gates

UPPER BOUND · witness at $m = 8$

$s8 = x1 \wedge x3$	$s9 = x0 \wedge x5$
$s10 = x4 \wedge x5$	$s11 = x1 \wedge s10$
$s12 = x7 \wedge s11$	$s13 = x2 \wedge s9$
$s14 = x4 \wedge x7$	$s15 = s10 \wedge s13$

- ✓ re-executed from scratch
8 gates, all targets realised
- ✓ Z3: $\forall x. c(x) = Mx$
whole input space, 0.22 s
- ✓ Lean 4 kernel
axioms: [propext]

$\Rightarrow$ 8 gates suffice

+

LOWER BOUND · refutation at $k = 7$

CNF encoding
1,010 vars · 17,636 clauses

Glucose 4.2 $\rightarrow$ UNSAT
0.17 s, emitting a DRAT proof

DRAT proof object
18,118 lines · 0.91 MB

drat-trim (third party) ✓
s VERIFIED in 0.24 s

proof sha256 95fb1b6f...

$\Rightarrow$ 7 gates impossible